

Teacher review and acceptance

SCIENCE 24 | A1 REACTION DETECTIVE | Prepared October 7, 2026

What you are reviewing

A fixed standalone v1.2.0 candidate: one worked example, four practice cases, and one independent transfer. Proposed use is optional Unit A formative practice. It carries no course grade or required-progress credit.

Teacher approval is requested for scientific accuracy, selected curriculum alignment, answer fairness, reading demands and instructional suitability. Approval here does not certify the entire course, accessibility, hosting, SCORM or Brightspace behavior.

Open and review

1. Extract the review ZIP. Open PLAY.html in your browser; it is self-contained and works without an Internet connection. If local HTML is restricted, use the game/ folder through your approved local/static host.
2. Play the worked example and all five learner tasks. Try a misconception, revise it, use Previous, and inspect your review. Avoid real student information while reviewing.
3. Compare your reasoning with the complete records and answer rules on pages 3-8. Mark the decisions and comments. Complete the checklist and version-specific acceptance on pages 9-10.

Identity and approval boundary

Version: 1.2.0. Candidate identity: SHA-256 of PLAY.html (full value below). Any change to science, answer rules, feedback or learner behavior requires affected-area re-review and a newly identified candidate.

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Important limits for learners

Work remains in memory. Reloading or closing the page clears the session. Previous preserves work only within that session. Print your review is available. No answers are sent to a teacher or LMS.

Structured choices are checked. Written explanations are submitted and self-reviewed; neither text length nor four checked rubric boxes establishes scientific correctness. Completion is not mastery.

Decision remains with the teacher

All decision fields are unreviewed. Type comments into the PDF and save a copy, or print and write on it. A typed acknowledgement is a review record, not a cryptographic digital signature. Return the PDF and any marked defects to Dean.

Curriculum and evidence boundaries

Teacher review focus | Evidence current as of October 7, 2026

Selected alignment - independently checked against the published program

General outcome 2, pp.31-32: evidence, reactant/product comparison and reaction classification. All cases support parts of this; W01 demonstrates decomposition.

General outcome 4, p.32: everyday acid-base applications, corrosion protection and combustion emissions. P03, P02 and P04 respectively address selected examples.

Analyzing/interpreting and communication, p.33: evidence-supported conclusions and explanation. Authored records rehearse parts of these skills; students do not perform a real experiment.

Primary source: Alberta Education, Science 14-24 Program of Studies (2003, updated 2014), Science 24 Unit A, printed/PDF pp.31-33.

<https://education.alberta.ca/media/159714/sc1424.pdf>

Mapping is a scoped interpretation for teacher confirmation. Formula naming, equation balancing, independent composition/decomposition assessment, real laboratory/WHMIS skills and the full range of emission controls are not established by this game.

Independent local technical checks

842 engine tests passed; 88 core browser assertions passed on exact embedded PLAY.html.

277 responsive assertions passed at 360, 390, 430, 768, 1024, 1440 and 1648 CSS pixels, including 390px touch emulation.

39 keyboard/stress assertions passed: full keyboard-only learner route, reduced motion, escaped text, long text, image fallback and CSS 200% zoom/reflow stress.

88 core browser assertions also passed through the actual canonical multi-file HTTP preview. All 11 supplied runtime file hashes match their hosted bytes.

Imported v1.2 learner runtime was not altered. Local validation harness adjustments: CommonJS package boundary; macOS keyboard typeahead for native select controls; screenshot timeout increased from 5 to 30 seconds.

Actual in-app-browser worked/first-practice layout inspected; local preview remains available.

8 offline smoke checks passed through an actual file:// launch of the self-contained PLAY.html, including evidence reveal and backward-state restoration.

Checks still separate from this teacher decision

Independent Science 24 teacher science, answer-fairness and instructional-suitability review.

Real phone and actual VoiceOver/NVDA route; native browser zoom; actual Safari/Firefox and print-dialog behavior.

Final lesson placement, optional/required progress decision, target host/CSP/iframe and Brightspace/SCORM integration.

Any persistent save or LMS reporting is a separate implementation and acceptance decision.

Acceptance questions needing particular scrutiny

Is corrosion the appropriate checked label rather than allowing composition? Is acid-carbonate / neutralization appropriate class terminology? Are exact evidence pairs fair? Does the uncertainty follow-up teach limits without rewarding avoidance? Are the mandatory rubric checks useful rather than mechanical?

Detailed scientific rationales and reference links: docs/SCIENCE_AND_SCOPE.md. Detailed raw results and test-method limits: independent-validation.json and evidence/. Teacher suitability and learning effectiveness remain unapproved.

W01 - Peroxide Record

Worked example

Learner situation and observations

A virtual laboratory record identifies hydrogen peroxide before a change. Afterwards, water and oxygen are identified. The mixture warms with no external heating.

Observation: Gas bubbles appear. | Energy: The mixture warms; no external heat is supplied.

All supplied records (including initially hidden records)

- A. Product identity [w01_e1]: Water and oxygen are identified after the change.
- B. Starting substance [w01_e2]: Hydrogen peroxide is identified before the change.
- C. Temperature record [w01_e3]: The mixture warms while no external heating is applied.
- D. Catalyst record [w01_e4]: The catalyst is unchanged after the reaction; it is not consumed in the overall change.

Checked answer and evidence rules

Conclusion: chemical. Reaction target: decomposition.

Demonstration only; not independent performance. No learner answer is checked in this round.

Model explanation - comparison, not automated prose grading

The identified substances change from hydrogen peroxide to water and oxygen. New substances support a chemical change. One compound forms simpler products, so the reaction is decomposition.

Case decision

Unreviewed

Corrections / alternative answers / level concerns

P01 - Cooling Solution

Guided investigation

Learner situation and observations

A white crystalline solid disappears into water. The container cools from 22 degrees C to 18 degrees C and a clear solution remains. What does that establish?

Appearance: The solid is no longer visible. | Temperature: 22 degrees C to 18 degrees C

All supplied records (including initially hidden records)

A. Recovery record [p01_e1]: After the water is removed in the supplied record, a solid is recovered.

B. Identity comparison [p01_e2]: The supplied identity analysis reports no new substances in this model. The recovered solid matches the starting solid.

C. pH record [p01_e3]: The record reports that the pH stayed the same.

D. Warm-water comparison [p01_e4]: The solid dissolves at a different rate in warmer water.

Checked answer and evidence rules

Conclusion: physical. Reaction target: none.

Accepted pairs: A + B. Exactly two selected revealed records are required.

Not enough evidence is accepted as an unfinished checkpoint only before the identity record has been seen. The free identity follow-up then requires a supported physical-change classification. Reopening does not erase previously seen identity evidence.

Model explanation - comparison, not automated prose grading

The recovery and identity records support a physical change in this supplied model. The original substance is recovered and no new substances are reported. Cooling alone was not enough to decide.

Hint ladder: 1. Separate the temperature observation from the question of substance identity. / 2. Recovery is useful, but the recovered material must also be identified. / 3. Pair the recovery record with the identity comparison.

Misconception feedback: The cooling and disappearance are observations. Neither alone establishes a new substance. Compare the recovery and identity records.

Case decision

Unreviewed

Corrections / alternative answers / level concerns

P02 - Rusting Tool

Build the evidence

Learner situation and observations

An iron-containing tool is left in moist air. A reddish-brown coating develops. In this record, the tool and all attached coating are weighed together; none has flaked away.

Surface: A reddish-brown coating develops. | Mass: The tool with its coating has gained mass.

All supplied records (including initially hidden records)

A. Coating analysis [p02_e1]: The coating contains iron and oxygen and has different properties from the starting metallic iron.

B. Atmosphere record [p02_e2]: The tool was exposed to air containing oxygen and to moisture.

C. Surface observation [p02_e3]: The coating is reddish-brown.

D. Location record [p02_e4]: The tool was stored in a garage.

Checked answer and evidence rules

Conclusion: chemical. Reaction target: corrosion.

Accepted pairs: A + B; A + C. Exactly two selected revealed records are required.

Which response best limits further corrosion? Accepted: Use a suitable barrier coating to limit air and moisture contact.

Other options: Polish it, then leave the metal unprotected.; Do nothing; only the colour has changed.

Model explanation - comparison, not automated prose grading

The coating has a different composition and properties from the starting iron, supporting a chemical change: corrosion. Oxygen from the surroundings contributes to the retained coating and the measured mass gain. A suitable barrier limits further contact with oxygen and moisture.

Hint ladder: 1. Colour is suggestive. Which record compares the coating with the original metal? / 2. Use the coating analysis, then connect it to the exposure conditions or surface observation.

Misconception feedback: Do not explain the coating as a colour change alone. Compare the coating analysis with the original metal.

Case decision

Unreviewed

Corrections / alternative answers / level concerns

P03 - Antacid Record

Reduced guidance

Learner situation and observations

In a virtual model, a calcium-carbonate antacid encounters an acidic solution. Review the supplied records to classify the change. This is not a dosing or laboratory activity.

Observation: Bubbles appear. | Acidity: The pH moves towards neutral.

All supplied records (including initially hidden records)

- A. Product record [p03_e1]: Carbon dioxide, water and a dissolved calcium salt are identified as products of the change.
- B. Reactant record [p03_e2]: The starting reactants are the acidic solution and calcium carbonate.
- C. Bubble observation [p03_e3]: Gas bubbles are visible during the change.
- D. Tablet appearance [p03_e4]: The tablet becomes smaller during the supplied record.

Checked answer and evidence rules

Conclusion: chemical. Reaction target: neutralization.

Accepted pairs: A + B; A + C; A + D. Exactly two selected revealed records are required.

Model explanation - comparison, not automated prose grading

The identified products differ from the starting reactants, so the change is chemical. Acid reacting with calcium carbonate is an acid-carbonate neutralization, producing a calcium salt, water and carbon dioxide. The pH trend supports reduced acidity; bubbles alone do not identify the reaction.

Hint ladder: 1. Classify using the identities of reactants and products, not the bubbles alone. / 2. The product record and the reactant record together identify the acid-carbonate pattern.

Misconception feedback: Use the product record to determine whether new substances formed. Bubbles and a pH trend do not identify the reaction by themselves.

Case decision

Unreviewed

Corrections / alternative answers / level concerns

P04 - Furnace Combustion

Independent practice

Learner situation and observations

A fictional natural-gas furnace supplies heat. Use its inlet and exhaust records to classify the reaction and connect a product to an environmental consequence.

Energy: Heat is released. | Observation: Light is produced in the burner.

All supplied records (including initially hidden records)

- A. Inlet record [p04_e1]: Methane and oxygen are supplied to the model burner.
- B. Exhaust record [p04_e2]: Carbon dioxide and water are identified as products in this complete-combustion model.
- C. Energy record [p04_e3]: The burner releases heat and visible light.
- D. Casing record [p04_e4]: A metal surface becomes hot during operation.

Checked answer and evidence rules

Conclusion: chemical. Reaction target: combustion.

Accepted pairs: A + B. Exactly two selected revealed records are required.

Which carbon-containing product links this model to greenhouse-gas emissions? Accepted: Carbon dioxide

Other options: Water vapour; Methane; Oxygen

Model explanation - comparison, not automated prose grading

Methane and oxygen react to form carbon dioxide and water. These new products support a chemical change classified as combustion. Energy is released. Carbon dioxide is the carbon-containing greenhouse-gas product identified in this model.

Hint ladder: 1. Use the inlet and exhaust identities together. Decide which substance is a product, not an input.

Misconception feedback: Compare the inlet and exhaust identities. Heat or light alone does not identify a reaction.

Case decision

Unreviewed

Corrections / alternative answers / level concerns

T01 - Boiling Water

Independent transfer

Learner situation and observations

Water is heated until bubbles form and water vapour is produced. The vapour is cooled and condenses back to liquid water. The supplied record identifies the substance before heating and after condensation as water (H₂O).

Before heating: Liquid water (H₂O) | During heating: Bubbles and water vapour | After cooling: Liquid water (H₂O)

Checked answer and evidence rules

Conclusion: physical. Reaction target: none.

Classify the change. Explain why the bubbles do or do not establish a chemical reaction, using the identity record. No hints, tokens, artwork or pre-submission model. First complete response is retained separately from corrections and replay.

Model explanation - comparison, not automated prose grading

This is a physical change. The record identifies water before heating and after condensation; the substance remains H₂O. The bubbles are associated with the change of state, not proof of a new substance.

Misconception feedback: Separate an observation from a conclusion. Ask what the before-and-after identity record establishes.

Case decision

Unreviewed

Corrections / alternative answers / level concerns

Teacher audit checklist

Play the game and compare its rules with your intended learners and assessment demands

Records, explanations and terminology are scientifically defensible.

Unreviewed

Selected Unit A alignment and exclusions are accurate for the intended use.

Unreviewed

Accepted evidence pairs and rejected alternatives are fair and sufficient.

Unreviewed

P01 limited-evidence checkpoint and identity follow-up are instructionally appropriate.

Unreviewed

Corrosion protection and combustion-emissions choices are suitable.

Unreviewed

Incorrect-answer feedback diagnoses reasoning without creating misconceptions.

Unreviewed

Scaffold fading, sentence starters and hint timing suit the learners.

Unreviewed

T01 is a useful narrow independent task; first response and corrections are distinguished.

Unreviewed

Explanation models/rubric help reasoning; automated prose correctness is never claimed.

Unreviewed

Reading level, visual clarity, navigation and student-facing session-loss notices are suitable.

Unreviewed

Fictional records do not invite hazardous real-world reproduction.

Unreviewed

Optional practice, no gradebook reporting and no reload save are acceptable for this use.

Unreviewed

Overall findings / classroom adjustments

Comments, accommodations or required changes

Version-specific teacher acceptance

Complete only after review | Proposed use: optional Science 24 Unit A practice

This record binds your decision to the candidate below. It does not authorize a different version, required course credit, gradebook reporting, production hosting or LMS release. Remaining technical/device checks must be resolved or explicitly accepted in the release decision.

Version 1.2.0 | PLAY.html SHA-256:

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Reviewer name

Teaching role / Science 24 experience

School / organization

Review date

Decision

Conditions / required changes / accessibility accommodations

Recommended lesson placement (teacher recommendation; integration pending)

Typed acknowledgement or handwritten signature

I reviewed the named candidate for science and instructional suitability within the stated scope. My conditions above must be reconciled before release. Any substantive revision returns affected areas for re-review.

Return this saved PDF to Dean. Decision fields are intentionally unreviewed; no automated process may complete this acceptance on the teacher's behalf.